Light

GCSE Computer Science – Programming Testing and Robustness (Set 1)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is the main purpose of defensive design in programming?

- ○ **A.** To make the program run faster
- ○ **B.** To ensure the program can handle unexpected or invalid inputs without crashing
- ○ **C.** To add more features to the program
- ○ **D.** To reduce the file size of the program

Q2 (1 mark)

What is input validation?

- ○ **A.** Checking that data entered by a user meets specified requirements before processing
- ○ **B.** Displaying error messages on screen
- ○ **C.** Saving user input to a file
- ○ **D.** Converting input from one data type to another

Q3 (1 mark)

What is authentication in the context of programming?

- ○ **A.** Encrypting all data in a program
- ○ **B.** Testing the program for errors
- ○ **C.** Confirming the identity of a user, typically through a username and password
- ○ **D.** Compiling the source code into machine code

Q4 (1 mark)

What is a syntax error?

- ○ **A.** An error that produces incorrect output but allows the program to run
- ○ **B.** An error caused by dividing by zero
- ○ **C.** An error that occurs when the program runs out of memory

○ **D.** An error that breaks the grammatical rules of the programming language and prevents the program from running

Q5 (1 mark)

What is a logic error?

○ **A.** An error that prevents the program from being translated

○ **B.** An error caused by a missing semicolon

○ **C.** An error in the program code that causes unexpected or incorrect output

○ **D.** An error that displays an error message to the user

Q6 (1 mark)

What is normal test data?

○ **A.** Data that should cause the program to crash

○ **B.** Data of the wrong data type

○ **C.** Data that is on the edge of being valid

○ **D.** Data that should be accepted by the program without causing errors

Q7 (1 mark)

What is boundary test data?

- ○ **A.** Data that is clearly invalid
- ○ **B.** Data of the correct type that is on the very edge of being valid
- ○ **C.** Data that should be accepted without any issues
- ○ **D.** Data that is of the wrong data type entirely

Q8 (1 mark)

Why is commenting code useful?

- ○ **A.** It helps other programmers understand what the code does
- ○ **B.** It reduces the file size of the program
- ○ **C.** It makes the program run faster
- ○ **D.** It prevents syntax errors from occurring

Q9 (1 mark)

Which variable name follows good naming conventions?

- ○ **A.** x
- ○ **B.** a1
- ○ **C.** ts
- ○ **D.** totalScore

Q10 (1 mark)

What is iterative testing?

- ○ **A.** Testing the complete program only once at the very end of development
- ○ **B.** Testing performed by the end users of the software
- ○ **C.** Testing modules of a program during development, fixing errors as they are found
- ○ **D.** Testing that only uses boundary data

Q11 (1 mark)

What is final (terminal) testing?

- ○ **A.** Testing performed by running the code in a terminal window
- ○ **B.** Testing each function separately during development
- ○ **C.** Testing that only checks for syntax errors
- ○ **D.** Testing the complete program at the end of production to ensure it meets all requirements

Q12 (1 mark)

Why is proper indentation important in programming?

- ○ **A.** It shows the structure of the code and makes it easier to read
- ○ **B.** It reduces the amount of memory used

- C. It makes the program execute faster
- D. It automatically fixes syntax errors

Q13 (2 marks)

A program accepts ages between 18 and 65 inclusive. Which of the following is an example of invalid test data?

- A. 18
- B. 40
- C. 'forty'
- D. 10

Q14 (2 marks)

A program expects the user to enter their age as a whole number. Which of the following is erroneous test data?

- A. 25
- B. -5
- C. 'twenty-five'
- D. 150

Q15 (2 marks)

A student writes the following code:

```
if score > 50
    print('Pass')
```

The code will not run. What type of error is present?

- ○ **A.** Syntax error
- ○ **B.** Runtime error
- ○ **C.** Logic error
- ○ **D.** Arithmetic error

Q16 (2 marks)

A program to calculate the average of three numbers contains the following code:

```
average = (num1 + num2 + num3) / 2
```

The program runs but gives incorrect results. What type of error is this?

- ○ **A.** Syntax error
- ○ **B.** Logic error
- ○ **C.** Runtime error
- ○ **D.** Input error

Q17 (2 marks)

A programmer wants to ensure a user enters a number between 1 and 10 inclusive. Which validation check would be most appropriate?

- ○ **A.** Length check
- ○ **B.** Presence check
- ○ **C.** Format check

- ○ **D.** Range check

Q18 (2 marks)

A login system checks a username and password. Why should passwords be stored as hashes rather than plain text?

- ○ **A.** If the database is compromised, attackers cannot see the actual passwords
- ○ **B.** Hashes make the program run faster
- ○ **C.** Hashes take up less storage space than plain text
- ○ **D.** Hashing automatically checks password strength

Q19 (2 marks)

A program asks users to enter a quantity of items to purchase. Why might a programmer add validation to reject negative numbers?

- ○ **A.** Negative numbers take up more memory
- ○ **B.** Negative numbers cannot be stored in variables
- ○ **C.** Negative numbers always cause syntax errors
- ○ **D.** Users might accidentally or deliberately enter negative values, which would be meaningless for quantity

Q20 (2 marks)

How does breaking a program into sub programs (functions and procedures) improve maintainability?

- ○ **A.** Code is organised into reusable sections that can be understood, tested, and modified independently
- ○ **B.** It makes the program use less memory
- ○ **C.** It prevents all errors from occurring
- ○ **D.** It removes the need for comments in the code

Q21 (2 marks)

A password must be between 8 and 20 characters long. Which values would be suitable boundary test data for the password length?

- ○ **A.** 5 and 25 characters
- ○ **B.** 8 and 20 characters
- ○ **C.** 10 and 15 characters
- ○ **D.** 1 and 100 characters

Q22 (2 marks)

What is the main purpose of testing a program?

- ○ **A.** To make the program run faster
- ○ **B.** To convert the program into machine code
- ○ **C.** To add new features to the program
- ○ **D.** To find and fix errors before the program is released to users

Q23 (2 marks)

Which information should be included in a test plan for a program?

- ○ **A.** Test data to be used, expected results, and actual results for each test
- ○ **B.** Only the names of the programmers who wrote the code
- ○ **C.** Only a list of features the program should have
- ○ **D.** Only the date when testing will be completed

Q24 (3 marks)

A program requires users to enter a UK postcode, which should start with one or two letters, followed by one or two digits.
Analyse which combination of validation checks would best ensure valid input.

- ○ **A.** Presence check only
- ○ **B.** Length check and range check
- ○ **C.** Presence check, length check, and format check to verify the pattern of letters and digits
- ○ **D.** Type check to ensure it is a string

Q25 (3 marks)

A program accepts exam marks from 0 to 100 inclusive. A tester needs to create a comprehensive test plan.
Analyse which set of test data would provide the most thorough testing.

- ○ **A.** 50, 75, 90 (normal data only)

- **B.** 0, 100 (boundary data only)
- **C.** Normal: 50; Boundary: 0, 1, 99, 100; Invalid: -1, 101; Erroneous: 'fifty'
- **D.** -10, 110, 'abc' (invalid and erroneous only)

Q26 (3 marks)

A program should display 'Adult' if age >= 18 and 'Child' otherwise. The code is:

```
if age > 18:
    print('Adult')
else:
    print('Child')
```

A user aged exactly 18 is shown 'Child'. Analyse the error and how to fix it.

- **A.** Syntax error - add a colon after the else statement
- **B.** Logic error - the condition should be >= 18, not > 18, so that 18-year-olds are classified as adults
- **C.** Runtime error - the variable age has not been defined
- **D.** No error exists - the program is working correctly

Q27 (3 marks)

A login system allows unlimited password attempts. An attacker tries thousands of common passwords automatically.
Analyse what defensive design measure would best prevent this attack.

- **A.** Requiring passwords to be at least 8 characters long

- ○ **B.** Limiting the number of failed login attempts before locking the account temporarily
- ○ **C.** Storing passwords in plain text for easier checking
- ○ **D.** Removing the password requirement entirely

Q28 (3 marks)

A programmer writes all their code in a single function with single-letter variable names and no comments.
Analyse the maintainability problems this creates.

- ○ **A.** The code will be difficult to understand, debug, and modify because the structure is unclear, variable purposes are unknown, and there is no documentation explaining the logic
- ○ **B.** The code will have syntax errors
- ○ **C.** No problems - shorter code runs faster
- ○ **D.** The program will crash when executed

Q29 (3 marks)

A development team discovers a major bug during final testing that traces back to a module written in the first week of a six-month project.
Analyse why iterative testing would have been beneficial.

- ○ **A.** Iterative testing is not beneficial - all testing should happen at the end
- ○ **B.** The bug would have been found early when the module was written, making it cheaper and easier to fix before other code depended on it
- ○ **C.** Iterative testing would have made the program run faster

- ○ **D.** Iterative testing would have automatically fixed the bug

Q30 (3 marks)

A booking system allows users to enter a date for an appointment. The system should reject past dates, weekends, and public holidays. Analyse the validation approach needed.

- ○ **A.** Multiple validation checks: verify it is a real date (format check), it is not in the past (comparison with current date), it is not a weekend (day-of-week check), and it is not a public holiday (lookup against a stored list)
- ○ **B.** A simple range check comparing the date to today's date
- ○ **C.** Only a presence check to ensure a date was entered
- ○ **D.** Convert the date to a number and check if it is positive

Total: **55** marks • Time: **30 min**

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